VU KHANH CAO

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EDUCATION

Carleton University, Ottawa, ON (GPA: 3.75/4.00) Sep 2023 – Apr 2028
Bachelor of Computer Science (Honours) – Artificial Intelligence and Machine Learning
Coursework: Object-Oriented Software Engineering, Distributed Computing, Design and Analysis of Algorithms

EXPERIENCE

Software Developer May 2026 – Present
Bank of Canada *Ottawa, ON*

- Modernized Treasury Management Systems processing over **$1 trillion in annual transactions** by developing Java Spring Boot services and TypeScript applications, and validating with Playwright.
- Built an agentic code-review suite integrated into Azure CI/CD pipelines to automatically enforce security and quality requirements across mission-critical systems, improving standards compliance by **60%**.
- Migrated legacy applications onto OpenShift-based cloud infrastructure, improving deployment reliability, scalability, and operational efficiency for systems supporting **2,000+ employees**.

Software Developer May 2025 – Dec 2025
Ovenns Analytics Inc. *Ottawa, ON*

- Developed a classification model using **logistic regression, SciPy, and NumPy** to categorize **3,000+ ingredients and preservatives** into food groups, achieving approximately **96% accuracy**.
- Built a **Node.js, TypeScript, React** pipeline that parsed online menus, normalized menu data, and decomposed compound ingredients into basic ingredients, supporting onboarding for **100+ clients**.

PROJECTS

GURT – GitHub Jan 2026

- Engineered a **real-time robotic control stack** spanning QNX 8, Raspberry Pi 4, Arduino motor control, and WebSocket communication, enabling live video streaming with **sub-100 ms command latency**.
- Developed a custom **QNX 8 camera driver** that increased video throughput from ∼**3 FPS to a stable 30 FPS (10×)**, helping earn **Best Hardware Hack** at uOttaHack 8.

JARVIS – GitHub uOttaHack - Jan 2025

- Built a low-cost **physical-to-digital reconstruction pipeline** using QNX, Raspberry Pi, and Gaussian Splatting to produce web-ready 3D scenes from webcams, reducing capture hardware cost from **$400 to $20**.
- Containerized the reconstruction pipeline with **Docker** and integrated **AWS S3 storage with GPU accelerated processing** to automate image ingestion, model training, and web delivery of generated 3D scenes.

TRACY – GitHub QHacks - Feb 2024

- Developed a **monocular computer vision pipeline** using TrackNetV2 and convolutional RNNs to reconstruct player and ball trajectories from a **single camera**, eliminating the need for specialized hardware.
- Built gameplay analytics for **serve speed, ball bounces, player heatmaps, and in/out detection** using Matplotlib and FFmpeg, contributing to a **3rd-place finish at QHacks 2024**.

SKILLS

Languages	Python, Java, JavaScript, TypeScript, C/C++, SQL
Libraries & Frameworks	React, Angular, Node.js, Express, Spring Boot, TensorFlow, OpenCV
Platforms & Technologies	Azure, AWS, Docker, Kubernetes, ArgoCD, OpenShift, REST APIs, Git